1- Each element of B is accessed N^3 times $= (32)^3 = 32768$

Naive DRAM traffic $= 2N^3 * 4 = 2(32)^3 * 4 = 262144$ bytes

2- The number of tiles in each dimension is $\left(\frac{N}{T}\right) = \left(\frac{32}{8}\right) = 4$

The number of tiles in the matrix $= 4 * 4 = 16$

The number of elements in each tile is $T^2 = (8)^2 = 64$ elements

The tile loads in A and B is 16 each

Elements in A and B is $64 * 16 = 1024$ element each

Total element is bytes $= (1024) * 2 * 4 = 8192$ bytes tiled DRAM traffic

3- Traffic Ratio $= \frac{\text{Naive DRAM traffic}}{\text{Tiled DRAM traffic}} = \frac{262144}{8192} = 32$

The naive multiplication rereads A and B for each of the dot product step while tiled multiplication brings elements from the DRAM once.

4- Total FLOP $= 2N^3 = 65536$

Compute time $= \frac{\text{Total FLOP}}{\text{DRAM compute}} = \frac{65536}{10 * 10^{12}} = 6.55$ ns

Naive memory time $= \frac{\text{Naive traffic}}{\text{DRAM bandwidth}} = \frac{262144}{320 \times 10^9} = 819.2$ ns

Tiled memory time $= \frac{\text{tiled DRAM traffic}}{\text{DRAM bandwidth}} = \frac{8192}{320 \times 10^9} = 25.6$ ns

For both naive and tiled the bottleneck is memory